Activity #2 – Operators

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BSIT 2-1 **Date: March 3, 2026**

General Instructions:

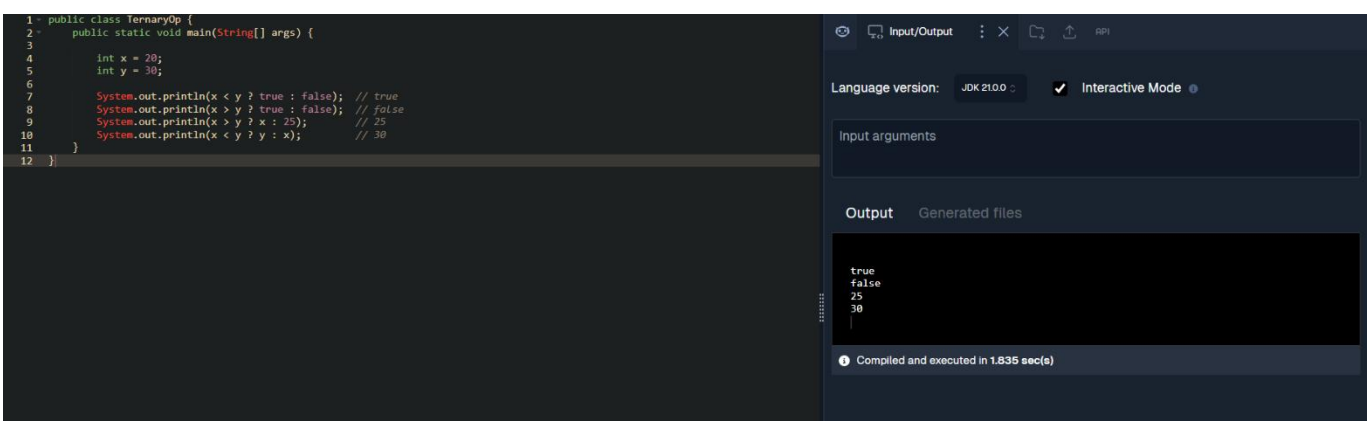
1. Create a Java program based on the questions.
2. Follow the exact required output shown in the question.
3. You must use the specific operator required in each number.
4. Do NOT hardcode the final answers (e.g., System.out.println(86); without computation is not allowed).
5. Show proper use of variables and correct operator usage.
6. Follow proper Java syntax, indentation, and naming conventions.
7. Use System.out.println() for displaying output.
8. Screenshot the code and the output.
9. Compile and run your program to verify correctness before submission.
10. Submission Format: Section_LastName_FirstName_OperatorsActivity#2.pdf

Please answer the following questions:

PART I:

1. Create a Java program using bitwise operators that will produce the output: true, 10, false, 15.

2. Create a Java program using the ternary operator that will produce the output: true, false, 25, 30.



3. Create a Java program using assignment operators that will produce the output: Addition: 120, Subtraction: 70, Multiplication: 144, Division: 16.

```java
public class AssignmentOp {
    public static void main(String[] args) {

        int val = 100;

        val += 20;
        System.out.println("Addition: " + val);   // 120

        val = 100;
        val -= 30;
        System.out.println("Subtraction: " + val);   // 70

        int mult = 12;
        mult *= 12;
        System.out.println("Multiplication: " + mult);   // 144

        int div = 64;
        div /= 4;
        System.out.println("Division: " + div);   // 16
    }
}
```

Input/Output

Language version: JDK 21.0.0 Interactive Mode

Input arguments

Output Generated files

```
Addition: 120
Subtraction: 70
Multiplication: 144
Division: 16
```

Compiled and executed in 1.964 sec(s)

Program Requirements:

1. Use at least five (5) different operators from the following categories:
   - Arithmetic operators (+, -, *, /, %)
   - Relational operators (>, <, >=, <=, ==, !=)
   - Logical operators (&&, ||, !)
   - Unary operators (++, --)
   - Assignment operators (+=, -=, *=, /=)
   - Bitwise operators (&, |, ^, <<, >>)

2. The program must produce exactly six (6) outputs only.
3. At least one output must include a tricky increment or decrement case, such as:
   - a++ + ++a
   - --b + b++
   - or any similar expression that modifies the variable within the same statement.
4. Do NOT reuse:
   - Any variable values
   - Any variable names
   - Any logic patterns from Part I.
5. The output must contain:
   - At least two (2) boolean results

- At least four (4) numeric results

```java
public class OperatorTask {
    public static void main(String[] args) {

        // Setup with unique variables and values
        int a = 8;
        int b = 3;
        double c = 25.5;
        boolean isLogged = false;

        // 1. Relational Operator (Boolean #1)
        System.out.println(a > b); // true

        // 2. Logical Operator (Boolean #2)
        System.out.println(!isLogged || (b == 0)); // true

        // 3. Arithmetic Operator (Numeric #1)
        System.out.println(c - 10.5); // 15.0

        // 4. Bitwise Operator (Numeric #2)
        System.out.println(a | b); //  11

        // 5. Unary Tricky Case (Numeric #3)
        int trickyResult = --a + a++;
        System.out.println(trickyResult); // 14

        // 6. Assignment Operator (Numeric #4)
        int d = 100;
        d /= 4;
        System.out.println(d); // 25
    }
}
```

Input/Output API

Language version: JDK 21.0.0 Interactive Mode

Input arguments

Output Generated files

```
true
true
15.0
11
14
25
```

Compiled and executed in **2.361 sec(s)**